

BATTULA KOTESH

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B.Tech, Electronics and Communication Engineering
Address : Neerukonda

Education

SRM University AP B.Tech · Electronics and Communication Engineering	2027 CGPA - 7.92/10
SRI SADHANA JUNIOR COLLEGE Class XII - BOARD OF INTERMEDIATE EDUCATION · MPC · MARKAPUR	2023 Percentage - 91.4%
APSWR SCHOOL FOR BOYS Class X - BOARD OF SECONDARY EDUCATION · DARSI	2021 Percentage - 85.8%

Experience

SRM University-AP Link Image processing filters using vivado HLS · Intern · Internship	May 2025 - Jul 2025 Amaravati, Andhra Pradesh
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Technologies: Vivado HLS 2018.3, C Programming, Image Processing, HLS Optimization, C/RTL Simulation

- Implemented Grayscale, Gaussian Blur, Laplacian, Sharpen, and Sobel image processing filters using C programming in Xilinx Vivado HLS.
- Developed testbenches and performed C simulation and RTL co-simulation to verify filter functionality and output correctness.
- Applied HLS optimization techniques including loop pipelining, array partitioning, and function inlining to improve hardware efficiency.
- Gained hands-on experience in Image Processing, Hardware Acceleration, **FPGA-Oriented** Design, Hardware Synthesis, and Verification.

Projects

Fire Fighter Robot Using Arduino Link SRM University-AP · Arduino, ESP32-CAM, ESP8266, Embedded C	Sept 2025 - Oct 2025 Embedded Systems & IoT
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Technologies: Arduino UNO, ESP32-S, ESP32-CAM, Arduino IDE, C Programming

- Developed an Arduino-based fire-fighting robot capable of detecting and extinguishing fire using flame sensors and a water pump.
- Integrated ESP32-S for Bluetooth-based remote control and ESP32-CAM for live video streaming and real-time monitoring.
- Implemented both manual and autonomous operating modes to improve safety and operational flexibility.
- Strengthened practical skills in Embedded Systems, IoT, Robotics, Sensor Interfacing, and Hardware Integration.

Automatic Night Light Control System Link SRM University-AP · Arduino, Arduino IDE, Circuit Design, Sensor Interfacing	Mar 2025 - Apr 2026 Embedded Systems & Electronics
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Technologies: Arduino Nano, LDR Sensor, Relay Module, Arduino IDE, C Programming

- Developed an automatic lighting system using an LDR sensor and Arduino Nano to monitor ambient light intensity.
- Designed and tested relay-based hardware for automatic ON/OFF switching of lights.
- Improved energy efficiency by reducing unnecessary power consumption through automated lighting control.
- Enhanced practical knowledge of Embedded Systems, Circuit Design, Sensor Interfacing, and Hardware Testing

Smart Accident Detection System Link SRM University-AP · Accelerometer, Arduino UNO, Embedded Systems, GSM Module, Sensor Interfacing	Mar 2025 - Apr 2026 Embedded Systems & IoT
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Technologies: Arduino UNO, GSM Module, Accelerometer, Arduino IDE, C Programming

- Designed and developed an Arduino-based accident detection system using an accelerometer and GSM module.
- Automated emergency alerts by detecting sudden vehicle impacts and sending notifications to predefined contacts.
- Integrated location-sharing functionality to support faster emergency response and improve road safety.
- Gained hands-on experience in Embedded Systems, IoT, Sensor Interfacing, and GSM Communication

Skills

: python, java, C language

Languages

Telugu [Native], English [Fluent], Hindi [Basic]

Links

[GitHub](#), [LeetCode](#), [LinkedIn](#)